

Q1: It is desired to find the optimal solution of the following optimization problem using the internal penalty function method and compare it with the result obtained from the graphical method.

$$\text{Minimize } f(x_1, x_2) = (x_1 - 1)^2 + (x_2 - 5)^2$$

Subject to

$$-x_1^2 + x_2 \leq 4$$

$$-(x_1 - 2)^2 + x_2 \leq 3$$

Solution

Interior penalty function method

The interior penalty function for the inequality constraints is

$$\phi(\mathbf{X}) = (x_1 - 1)^2 + (x_2 - 5)^2 - \frac{1}{r} \left(\frac{1}{-x_1^2 + x_2 - 4} + \frac{1}{-(x_1 - 2)^2 + x_2 - 3} \right)$$

$$\nabla \phi(\mathbf{X}) = \begin{bmatrix} 2(x_1 - 1) + \frac{2x_1}{r(-x_1^2 + x_2 - 4)^2} + \frac{2(x_1 - 2)}{r(-(x_1 - 2)^2 + x_2 - 3)^2} \\ 2(x_2 - 5) - \frac{1}{r(-x_1^2 + x_2 - 4)} - \frac{1}{r(-(x_1 - 2)^2 + x_2 - 3)} \end{bmatrix}$$

The gradient of ϕ for unconstrained minimization is

$$\begin{aligned} \frac{\partial \phi}{\partial x_1} &= 2(x_1 - 1) + \frac{2x_1}{r(-x_1^2 + x_2 - 4)^2} + \frac{2(x_1 - 2)}{r(-(x_1 - 2)^2 + x_2 - 3)^2} \\ \frac{\partial \phi}{\partial x_2} &= 2(x_2 - 5) - \frac{1}{r(-x_1^2 + x_2 - 4)} - \frac{1}{r(-(x_1 - 2)^2 + x_2 - 3)} \end{aligned}$$

For $r \rightarrow 0$ and $\nabla \phi = 0$,

$$\begin{aligned} x_1 &= 1 \\ x_2 &= 5 \end{aligned}$$

This answer is not feasible. We start from a feasible $\mathbf{X}$, choose a value of r and decrease it.

A feasible starting point is $\mathbf{X}^{(0)} = (0, 0)$, $g_1 = -4$, $g_2 = -7$. The penalty parameter is reduced sequentially

$$r_k = \frac{1}{k}$$

Iteration 1 ($k = 1$, $r_1 = 1$):

$$\begin{aligned} \frac{\partial \phi}{\partial x_1} &= 2(x_1 - 1) + \frac{2x_1}{-x_1^2 + x_2 - 4} + \frac{2(x_1 - 2)}{-(x_1 - 2)^2 + x_2 - 3} \\ \frac{\partial \phi}{\partial x_2} &= 2(x_2 - 5) - \frac{1}{-x_1^2 + x_2 - 4} - \frac{1}{-(x_1 - 2)^2 + x_2 - 3} \end{aligned}$$

Starting from $\mathbf{X}^{(0)} = (0, 0)$, we minimize $\phi(\mathbf{X}, 1)$ using a gradient-based unconstrained method. The necessary conditions $\nabla \phi = 0$ lead to the system

$$\begin{aligned} 2(x_1 - 1) + \frac{2x_1}{-x_1^2 + x_2 - 4} + \frac{2(x_1 - 2)}{-(x_1 - 2)^2 + x_2 - 3} &= 0 \\ 2(x_2 - 5) - \frac{1}{-x_1^2 + x_2 - 4} - \frac{1}{-(x_1 - 2)^2 + x_2 - 3} &= 0 \end{aligned}$$

Solving numerically yields

Check: (feasible).

Iteration 2 ($k = 2, r_2 = 0.1$):

Using $\mathbf{X}_1^*$ as the starting point and minimizing $\phi(\mathbf{X}, 0.1)$ gives

with . The process continues for decreasing r until convergence.

Sequential minimization was carried out for $r \in \{1, 0.1, 0.01, 0.001, 0.0001\}$. The table below lists the obtained optimum $\mathbf{X}^*(r)$ together with the objective and constraint values.

Table 1: Convergence of the interior penalty method

k	r	x_1^*	x_2^*	$f(\mathbf{X}^*)$	$g_1(\mathbf{X}^*)$	$g_2(\mathbf{X}^*)$
1	1.0	0.8200	4.1500	0.7550	-0.5224	-0.2424
2	0.1	0.7520	4.5550	0.2595	-0.0105	-0.0025
3	0.01	0.7502	4.5619	0.2539	-1.1e-4	-1.0e-4
4	0.001	0.75002	4.56249	0.253906	-1.2e-5	-1.1e-5
5	0.0001	0.75000	4.56250	0.253906	-1.2e-6	-1.1e-6
...	$r \rightarrow 0$	0.75	4.5625	0.25390625	0	0

The exact optimum is the intersection of the two constraint boundaries, found from $g_1 = g_2 = 0$, which gives $x_1 = 0.75$, $x_2 = 4.5625$, and agrees with the limit of the penalty method as $r \rightarrow 0$.

$$x_1^* = 0.75, \quad x_2^* = 4.5625, \quad f(\mathbf{X}^*) = 0.2539$$

Graphical Solution

According to the reference notes, we solve this problem in six steps.

Step 1: Create a suitable coordinate system

Draw the x_1 - x_2 plane with x_1 on the horizontal axis and x_2 on the vertical axis.

Step 2: Draw the constraint boundaries

- g_1 : $-x_1^2 + x_2 = 4 \Rightarrow x_2 = x_1^2 + 4$. This is an upward-opening parabola with vertex at $(0, 4)$.
- g_2 : $-(x_1 - 2)^2 + x_2 = 3 \Rightarrow x_2 = (x_1 - 2)^2 + 3$. This is an upward-opening parabola with vertex at $(2, 3)$.

Step 3: Identify the feasible region for each constraint

- $g_1 \leq 0$: $-x_1^2 + x_2 - 4 \leq 0 \Rightarrow x_2 \leq x_1^2 + 4$. Feasible side is on or below the first parabola.
- $g_2 \leq 0$: $-(x_1 - 2)^2 + x_2 - 3 \leq 0 \Rightarrow x_2 \leq (x_1 - 2)^2 + 3$. Feasible side is on or below the second parabola.

Step 4: Determine the overall feasible region

The feasible region is the intersection of the half-planes below both parabolas. The two curves intersect where



hence $x_1 = 0.75$. At this point, $x_2 = (0.75)^2 + 4 = 4.5625$. Therefore the intersection point is $(0.75, 4.5625)$.

The feasible set is the region below the lower envelope of the two parabolas:

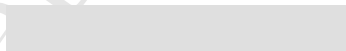
- For $x_1 < 0.75$ the lower curve is $x_2 = (x_1 - 2)^2 + 3$.
- For $x_1 > 0.75$ the lower curve is $x_2 = x_1^2 + 4$.

Step 5: Draw the objective function contours

$f(x_1, x_2) = (x_1 - 1)^2 + (x_2 - 5)^2 = C$ describes circles centered at $(1, 5)$ with radius $\sqrt{C}$. The function decreases as the circles shrink toward the center. Minimising f is equivalent to finding the smallest circle that still touches the feasible region.

Step 6: Identify the optimal design

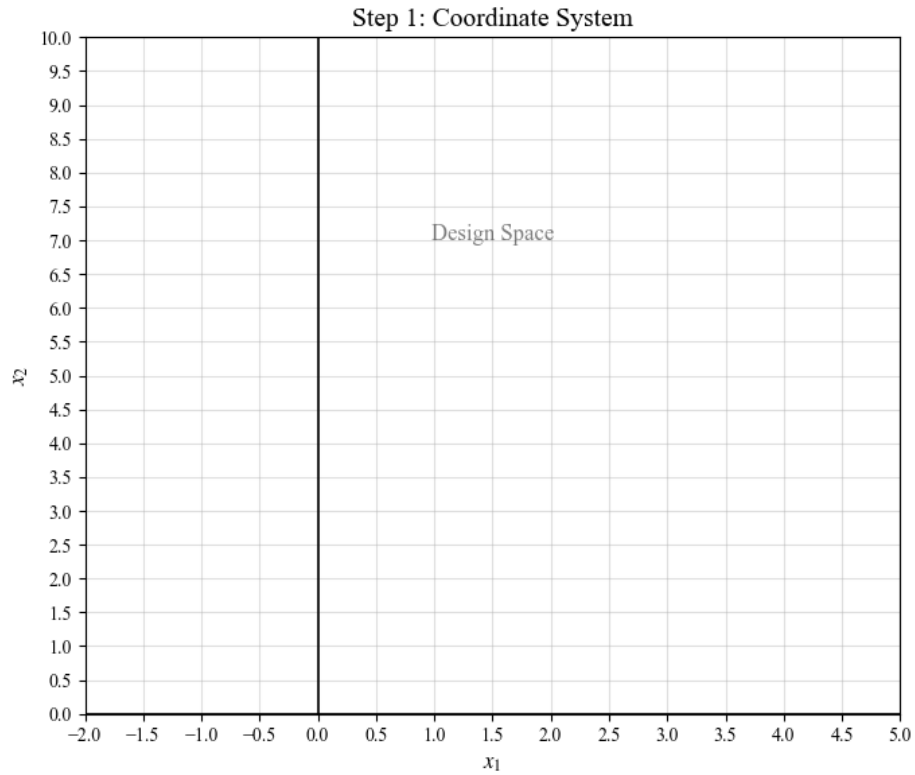
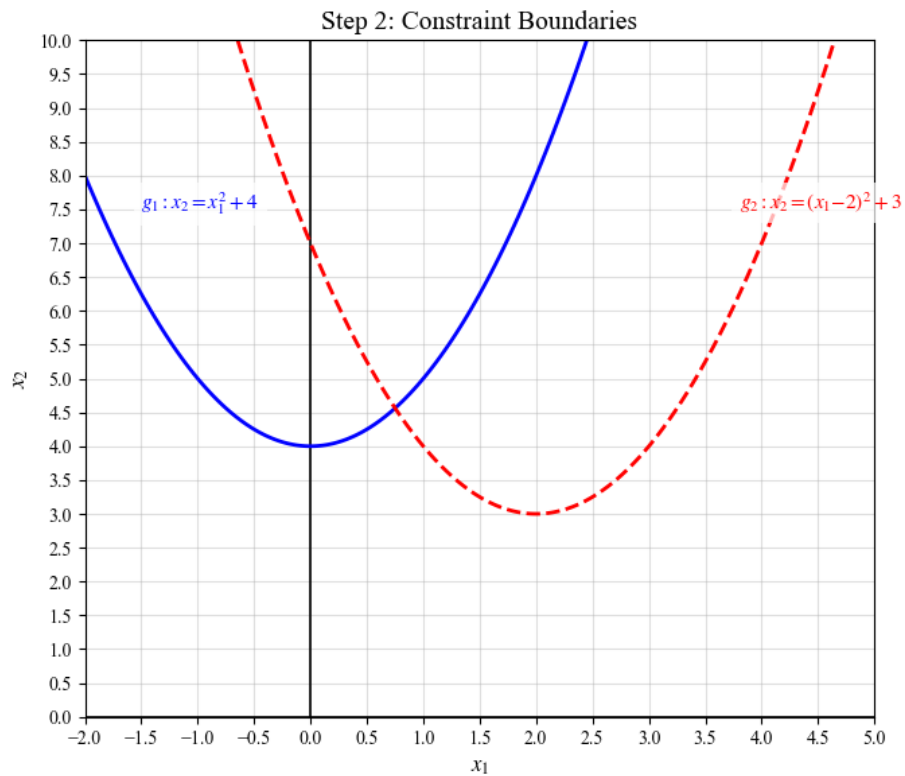
The center $(1, 5)$ is not feasible: at $x_1 = 1$ the upper bound is $\min(1^2 + 4 = 5, (1 - 2)^2 + 3 = 4) = 4$, and $5 > 4$. The optimal feasible point is the one on the boundary that minimises the distance to $(1, 5)$. That point is the intersection of the two constraint boundaries, because moving away from it along either parabola increases the distance. Hence,



and the minimum objective value is

$$x_1^* = 0.75, \quad x_2^* = 4.5625, \quad f(\mathbf{X}^*) = 0.2539$$

Results from both methods are the same; there is only a difference in the floating-point numbers.

**Figure 1:** The suitable coordinate system**Figure 2:** The constraint boundaries

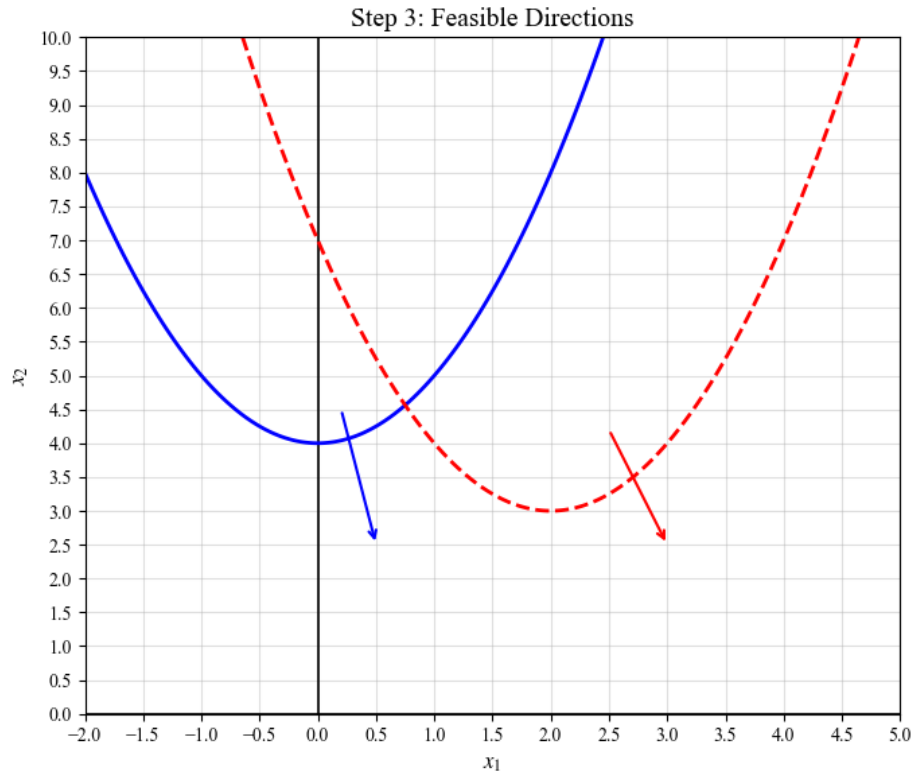


Figure 3: The feasible region for each constraint

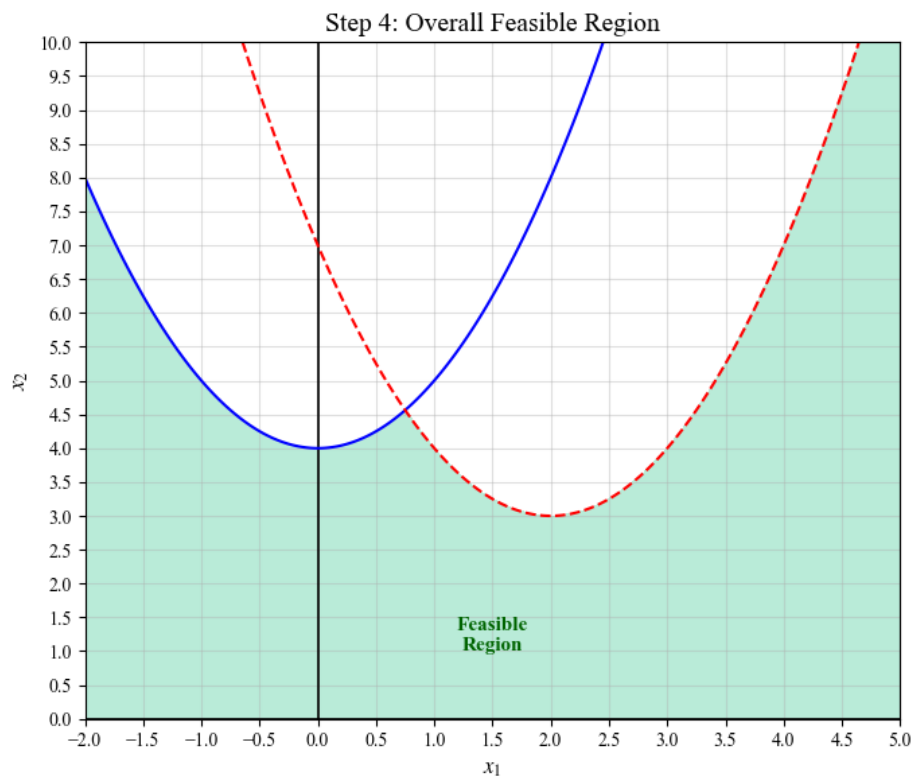


Figure 4: The overall feasible region

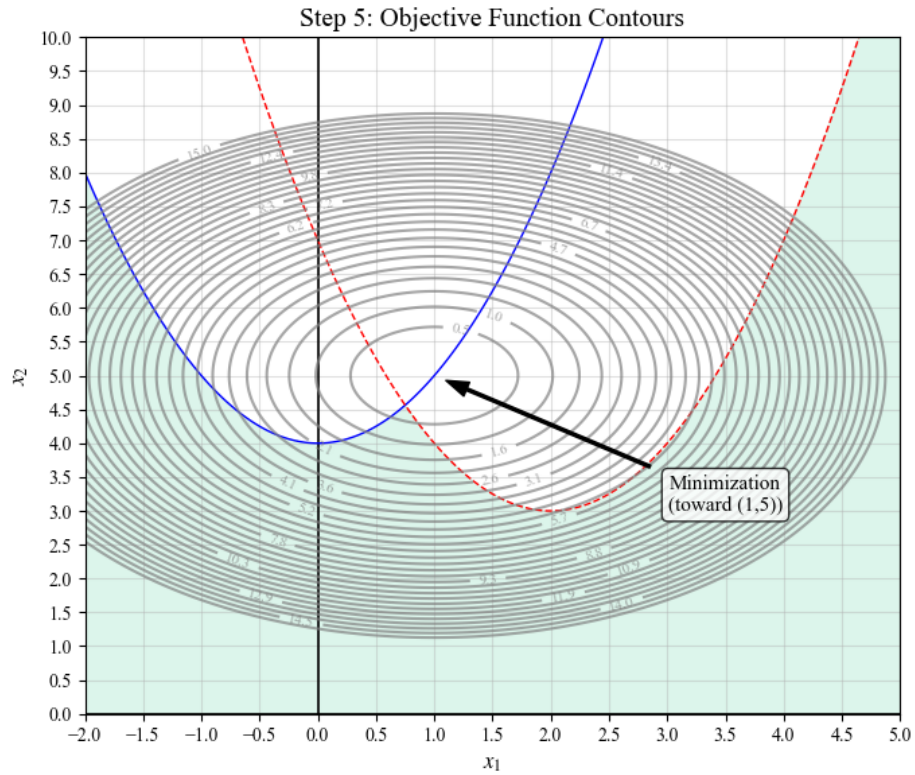


Figure 5: The objective function contours

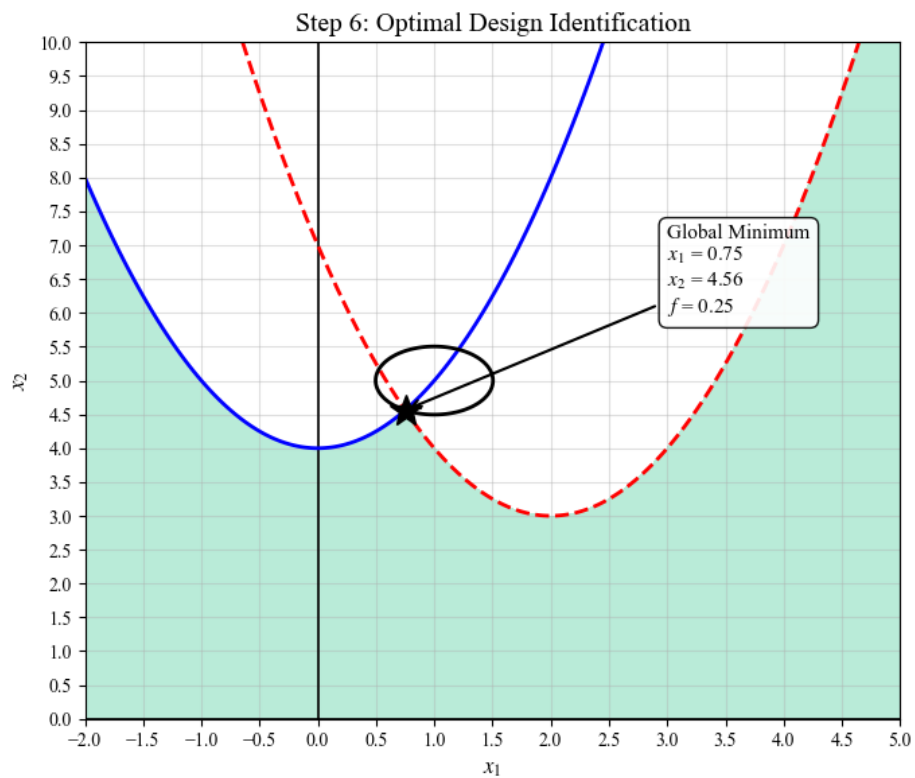


Figure 6: The optimal design

Q2: It is desired to find the optimal solution of the following optimization problem using the external penalty function method and compare it with the result obtained from the KKT method.

$$\begin{aligned} &\text{Minimize } f(x_1, x_2) = (2x_1 - x_2)^2 + (x_2 + 1)^2 \\ &\text{Subject to} \\ &\quad x_1 + x_2 = 10 \end{aligned}$$

Solution

Exterior Penalty Function Method

For an equality constrained problem, the exterior penalty function is formulated as

$$\phi(\mathbf{X}, r_k) = f(\mathbf{X}) + r_k [h(\mathbf{X})]^2$$

where $h(\mathbf{X}) = x_1 + x_2 - 10 = 0$ and r_k is a positive penalty parameter that increases with each iteration: $r_{k+1} = c r_k$, $c > 1$. As $r_k \rightarrow \infty$, the solution of the penalized problem approaches the true constrained optimum from the infeasible side.

The algorithm steps are

1. Choose an initial point $\mathbf{X}_1$ and an initial penalty parameter $r_1 > 0$. Set $k = 1$.
2. For fixed r_k , minimize $\phi(\mathbf{X}, r_k)$ using an unconstrained method to obtain $\mathbf{X}_k^*$.
3. Check convergence. If optimal, stop; otherwise, go to step 4.
4. Update $r_{k+1} = c r_k$ with $c > 1$.
5. Set $k = k + 1$, use $\mathbf{X} = \mathbf{X}_k^*$ as new starting point, and go to step 2.

The penalized objective is

$$\phi(x_1, x_2, r_k) = (2x_1 - x_2)^2 + (x_2 + 1)^2 + r_k (x_1 + x_2 - 10)^2$$

Setting the gradient to zero gives the linear system

$$\begin{aligned} \frac{\partial \phi}{\partial x_1} &= 4x_1 - 2x_2 + 2r_k(x_1 + x_2 - 10) = 0 \\ \frac{\partial \phi}{\partial x_2} &= -2x_1 + 2x_2 + 2r_k(x_1 + x_2 - 10) = 0 \end{aligned}$$

Solving these two linear equations yields the minimizer $\mathbf{X}^*(r) = (x_1^*(r), x_2^*(r))$ for any given r .

We choose the initial point $\mathbf{X}_1 = (0, 0)$, initial penalty $r_1 = 1$, and scaling factor $c = 10$. Iteration 1 ($k = 1, r_1 = 1$):

$$\begin{aligned} x_1^*(1) &= 10 \\ x_2^*(1) &= 0 \end{aligned}$$

Solving numerically yields

Iteration 2 ($k = 2, r_2 = 10$):

Sequential minimization was carried out for $r \in \{1, 10, 100, 1000, \dots\}$. The table below lists the obtained optimum $\mathbf{X}^*(r_k)$ together with the objective and constraint values.

Table 2: Convergence of exterior penalty method

Iter. (k)	r_k	x_1^*	x_2^*	$f(\mathbf{X}^*)$	$h(\mathbf{X}^*)$
1	1	2.785714	3.928571	52.306122	-3.285714
2	10	3.923077	5.634615	53.020408	-0.442308
3	100	4.082153	5.873515	52.904564	-0.044332
4	1000	4.098224	5.897337	52.900049	-0.004439
5	10000	4.099822	5.899734	52.900005	-0.000444
6	100000	4.099982	5.899973	52.900000	-0.000044
7	1000000	4.099998	5.899997	52.900000	-0.000004
...	$\rightarrow \infty$	4.1	5.9	52.9	0

$$x_1^* = 4.1, \quad x_2^* = 5.9, \quad f(\mathbf{X}^*) = 52.9$$

KKT Method Solution

According to the KKT Conditions, the solution follows six steps.

Step 1: Form the Lagrangian Function

The problem has one equality constraint $h(x) = x_1 + x_2 - 10 = 0$. Introduce a Lagrange multiplier v for the equality. The Lagrangian is

Step 2: Gradient Conditions

$$\frac{\partial L}{\partial x_1} = 0 \quad (1)$$

$$\frac{\partial L}{\partial x_2} = 0 \quad (2)$$

$$\frac{\partial L}{\partial v} = 0 \quad (3)$$

Step 3: Check Feasible Space

Equation (3) directly enforces the equality constraint $x_1 + x_2 = 10$.

Step 4: Check Switching Conditions There are no inequality constraints, so complementary slackness conditions do not apply.

Step 5: Check Non-negativity of Lagrangian Multipliers

The multiplier v corresponds to an equality constraint and is unrestricted in sign; no non-negativity condition is imposed.

Solving the KKT system From (3), $x_2 = 10 - x_1$. Substitute into (1) and (2),

$$(1) \Rightarrow \text{[redacted]}$$

$$(2) \Rightarrow \text{[redacted]}$$

Equating the expressions for v ,

$$\text{[redacted]}$$

Then [redacted] and [redacted].

The unique KKT point is $x^* = (4.1, 5.9)$ with multiplier $v^* = -9.2$

Step 6: Check Regularity of Candidate Points

The only constraint is $h(x) = x_1 + x_2 - 10 = 0$. Its gradient is

$$\text{[redacted]}$$

so the point is regular and the KKT conditions hold.

Optimal solution

$$x_1^* = 4.1, \quad x_2^* = 5.9, \quad f^* = 52.9$$

Results obtained from both methods are the same.

Q3: It is desired to find the optimal point in the following problem using the steepest descent method.

$$\text{Minimize } f(x_1, x_2) = x_1^2 + x_2^2 - 2x_1 - 4x_2 + 5$$

Solution

For solve this problem, use the following algorithm mentioned in slides.

1. Choose a starting point $\mathbf{X}^{(0)}$. Set $k = 0$.
2. Compute the gradient $\nabla f(\mathbf{X}^{(k)})$.
3. Convergence check: If $\|\nabla f(\mathbf{X}^{(k)})\| \leq \epsilon$, stop; $\mathbf{X}^{(k)}$ is optimal.
4. Set search direction $\mathbf{d}^{(k)} = -\nabla f(\mathbf{X}^{(k)})$.
5. Find the optimal step size λ_k^* that minimizes $f(\mathbf{X}^{(k)} + \lambda \mathbf{d}^{(k)})$.
6. Update the design: $\mathbf{X}^{(k+1)} = \mathbf{X}^{(k)} + \lambda_k^* \mathbf{d}^{(k)}$, set $k = k + 1$ and go to step 2.

For initial point, we use

[Redacted]

Iteration 1

[Redacted]

At $\mathbf{X}^{(0)}$:

[Redacted]

Search direction

[Redacted]

Exact line search: minimise f along $\mathbf{X}^{(0)} + \lambda \mathbf{d}^{(0)}$

[Redacted]

[Redacted]

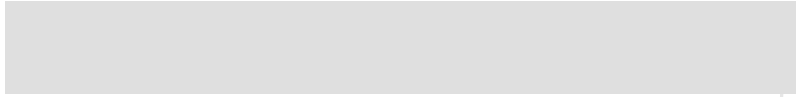
Update,

[Redacted]

Function value:



Iteration 2



The algorithm stops. The optimum is found in one iteration.

Table 3: Iterations of the steepest descent method

k	x_1	x_2	$f(\mathbf{X})$	$\ \nabla f\ $	λ^*
0	0.0000	0.0000	5.0000	4.4721	—
1	1.0000	2.0000	0.0000	0.0000	0.5000

Optimal solution

$$x_1^* = 1, \quad x_2^* = 2, \quad f^* = 0$$